United States Patent 12403732
Kind Code B1
Date of Patent September 02, 2025
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Garbage can hitch and garbage can having a built-in hitch

Abstract

A garbage can hitch includes a base section, and a receptacle disposed within the base section. The receptacle receives a hitch ball of a vehicle. The garbage can hitch includes a semi-cylindrical well disposed within the base section, which receives a handle bar of the garbage can. The garbage can hitch includes a first arm that extends from the base section, which can be positioned adjacent a first side of center cross stiffener of the garbage can. The garbage can hitch includes a second arm that extends from the base section, which can be positioned adjacent a second side of the center cross stiffener of the garbage can. A first width-adjusting clamp pad can be removably attached to the first arm. A second width-adjusting clamp pad can be removably attached to the second arm. The width-adjusting clamp pads can be clamped against the center cross stiffener of the garbage can.

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Appl. No.: **17/842707**

Filed: **June 16, 2022**

Related U.S. Application Data

continuation-in-part parent-doc US 17116899 20201209 US 11390127 child-doc US 17842707 us-provisional-application US 62954484 20191228

Publication Classification

Int. Cl.: **B60D1/07** (20060101); **B60D1/00** (20060101)

U.S. Cl.:

CPC **B60D1/07** (20130101); B60D2001/005 (20130101)

Field of Classification Search

CPC: B65F (1/468); B65F (1/1473); B60D (1/06); B60D (1/065); B60D (2001/001); B62B (5/0079)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6725855	12/2003	Brennan	N/A	N/A
7614637	12/2008	Kidd	280/480	B60D 1/00
8550318	12/2012	Tortumlu	280/480	B65F 1/1468
8998555	12/2014	Ingham	N/A	N/A
9586637	12/2016	Kentner	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2430422	12/2006	GB	B60D 1/00

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Background/Summary

RELATED APPLICATION DATA (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/116,899, filed on Dec. 9, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/954,484, filed on Dec. 28, 2019, which are hereby incorporated by reference.

TECHNICAL FIELD
(1) This application pertains to hitch devices, and more particularly, to a garbage can hitch for facilitating the movement of a garbage can using a vehicle, and a garbage can having a built-in hitch.
BACKGROUND
(2) Some people live in a home with a large set back from a main road, such that the garbage must be pulled along a driveway, which in some cases can extend for hundreds of feet. Usually in such circumstances, the driveway is made of gravel or dirt, making it even more difficult to drag the garbage can, particularly when the garbage can is full of garbage. Inclement weather can also increase the human misery of manually pulling a garbage can full of garbage up or down a long, muddy road.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1**A through **1**F illustrate a single-part garbage can hitch from various view angles in accordance with various embodiments of the present inventive concept.
(2) FIG. **2** illustrates a close-up side view of the single-part garbage can hitch of FIGS. **1**A through **1**F in accordance with various embodiments of the present inventive concept.
(3) FIG. **3** illustrates a close-up side view of the single-part garbage can hitch of FIGS. **1**A through **1**F with a handle bar of a garbage can disposed in a semi-cylindrical well in accordance with various embodiments of the present inventive concept.
(4) FIG. **4** illustrates a perspective view of the single-part garbage can hitch of FIGS. **1**A through **1**F being coupled with a garbage can in accordance with various embodiments of the present inventive concept.
(5) FIG. **5** illustrates a plan view of the single-part garbage can hitch of FIGS. **1**A through **1**F being coupled with a trailer hitch ball of a vehicle in accordance with various embodiments of the present inventive concept.
(6) FIG. **6** illustrates a side view of the single-part garbage can hitch of FIGS. **1**A through **1**F coupling the garbage can with the trailer hitch ball of the vehicle in accordance with various embodiments of the present inventive concept.
(7) FIG. **7** illustrates a perspective view of a garbage can having a built-in hitch for coupling the garbage can with the trailer hitch ball of the vehicle in accordance with various embodiments of the present inventive concept.
(8) FIG. **8** illustrates a plan view of the garbage can having the built-in hitch for coupling the garbage can with the trailer hitch ball of the vehicle in accordance with various embodiments of the present inventive concept.
(9) FIG. **9** illustrates a side view of the garbage can having the built-in hitch for coupling the garbage can with the trailer hitch ball of the vehicle in accordance with various embodiments of the present inventive concept.
(10) FIG. **10** illustrates another side view of the garbage can having the built-in hitch for coupling the garbage can with the trailer hitch ball of the vehicle in accordance with various embodiments of the present inventive concept.
(11) FIG. **11** illustrates yet another side view of the garbage can having the built-in hitch for coupling the garbage can with the trailer hitch ball of the vehicle in accordance with various embodiments of the present inventive concept.
(12) FIG. **12** is a flow diagram illustrating a technique for using a single-part garbage can hitch to tow a garbage can using a vehicle.
(13) FIG. **13** is a flow diagram illustrating another technique for using a single-part garbage can hitch to tow a garbage can using a vehicle.
(14) FIG. **14** is a flow diagram illustrating a technique for manufacturing a single-part garbage can hitch to tow a garbage can using a vehicle.
(15) FIG. **15** is a flow diagram illustrating a technique for manufacturing a garbage can having a built-in hitch to tow the garbage can using a vehicle.
(16) FIG. **16** is a perspective view of a garbage can hitch attached to a garbage can in accordance with various embodiments of the present inventive concept.
(17) FIG. **17** is a different perspective view of the garbage can hitch attached to the garbage can in accordance with various embodiments of the present inventive concept.
(18) FIG. **18** is another perspective view of the garbage can hitch attached to the garbage can in accordance with various embodiments of the present inventive concept.
(19) FIG. **19** is a close-up view of the garbage can hitch attached to the garbage can in accordance with various embodiments of the present inventive concept.
(20) FIG. **20** is another close-up view of the garbage can hitch attached to the garbage can in

accordance with various embodiments of the present inventive concept.
(21) FIG. **21** is a perspective view of the garbage can hitch including various different width-adjusting clamp pads in accordance with various embodiments of the present inventive concept.
(22) FIG. **22** is a schematic diagram of side views of the various different width-adjusting clamp pads in accordance with various embodiments of the present inventive concept.
(23) FIG. **23** is a bottom view of the garbage can hitch including the receptacle in accordance with various embodiments of the present inventive concept.
(24) FIG. **24** is a top view of the garbage can hitch in accordance with various embodiments of the present inventive concept.
(25) FIG. **25** is a first side view of the garbage can hitch in accordance with various embodiments of the present inventive concept.
(26) FIG. **26** is a second side view of the garbage can hitch in accordance with various embodiments of the present inventive concept.
(27) The foregoing and other features of the inventive concept will become more readily apparent from the following detailed description, which proceeds with reference to the accompanying drawings.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(28) Reference will now be made in detail to embodiments of the inventive concept, examples of which are illustrated in the accompanying drawings. The accompanying drawings are not necessarily drawn to scale. In the following detailed description, numerous specific details are set forth to enable a thorough understanding of the inventive concept. It should be understood, however, that persons having ordinary skill in the art may practice the inventive concept without these specific details. In other instances, well-known methods, procedures, components, circuits, and networks have not been described in detail so as not to unnecessarily obscure aspects of the embodiments.
(29) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first block could be termed a second block, and, similarly, a second block could be termed a first block, without departing from the scope of the inventive concept.
(30) It will be understood that when an element or layer is referred to as being “on,” “coupled to,” or “connected to” another element or layer, it can be directly on, directly coupled to or directly connected to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly coupled to,” or “directly connected to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.
(31) The terminology used in the description of the inventive concept herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the inventive concept. As used in the description of the inventive concept and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
(32) FIGS. **1**A through **1**F illustrate a single-part garbage can hitch **100** from various view angles in accordance with various embodiments of the present inventive concept. The single-part garbage can hitch **100** can be made of a single contiguous molded part. In an alternate embodiment, the

single-part garbage can hitch **100** can be made of multiple separate parts that are combined (e.g., attached to each other) to form a single functional part. The single-part garbage can hitch **100** is preferably made of plastic because it is lightweight and strong, but it will be understood that the single part garbage can hitch **100** can be made of any suitable hard material such as metal, fiber glass, wood, composite materials, or the like. In some embodiments, the single-part garbage can hitch **100** is manufactured using a plastic injection mold. In some embodiments, the single-part garbage can hitch **100** is manufactured using a 3D printer.

(33) As shown in FIG. **1**A, the single-part garbage can hitch **100** can include a grab handle **110**. The grab handle **100** has proportions such that any typical human hand can easily grasp the single-part garbage can hitch **100** using the grab handle **110**. The single-part garbage can hitch **100** can include a base section **105**. In some embodiments, the base section **105** is substantially in the form of a block. However, it will be understood that the base section **105** can have other suitable shapes without departing from the inventive aspects described herein.

(34) The single-part garbage can hitch **100** can include an attachment arm **120**, which can be used to attach to a center cross stiffener beam of a garbage can, as further described below. The attachment arm **120** can attach to multiple standard cross stiffener beam sizes. The attachment arm **120** can include a recess **130** located at an underside thereof. The single-part garbage can hitch **100** can include a semi-cylindrical well **115** located adjacent the attachment arm **120**, and between the attachment arm **120** and the grab handle **110**. The semi-cylindrical well **115** can be filled by a handle bar of a garbage can as further described below.

(35) As shown in FIG. **1**C, the single-part garbage can hitch **100** can include a receptacle **125** having a downwardly facing opening. The receptacle **125** can be located within the base section **105** of the single-part garbage can hitch **100**. The receptacle **125** can be shaped to receive a trailer hitch ball of a vehicle, as further described below.

(36) FIG. **2** illustrates a close-up side view of the single-part garbage can hitch **100** of FIGS. **1**A through **1**F in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **2** are described above, and therefore, a detailed description of these elements is not necessarily repeated. The receptacle **125** can include a hollow cylindrical section **205** and a hollow conical frustum section **215** having a tapered wall **210**. An end **225** of the conical frustum section **215** may meet at an end **230** of the cylindrical section **205**, thereby forming a contiguous open receptacle **125** within the base section **105**. The cylindrical section **205** can function to engage with a surface **235** of a hitch ball **220**. The frustum section **215** can function to engage with the surface **235** of the hitch ball **220**. For example, the tapered wall can function to keep the hitch ball **220** substantially centered within the receptacle **125**. The receptacle **125** can fit over any standard sized trailer hitch ball (e.g., **220**), such as 1⅞ inch, 2 inch, and 2 5/16 inch hitch balls. It will be understood that in alternate embodiments, the receptacle **125** can take other shapes that are suitable for being fit over any standard sized trailer hitch ball, without departing from the inventive aspects disclosed herein. For example, the receptacle **125** can have a hemispherical shape to receive the trailer hitch ball **220**. In some embodiments, the receptacle **125** has a box shape, a hexagonal shape, a triangular shape, or the like.

(37) The single-part garbage can hitch **100** enables an easy on-off motion for smooth and efficient usability. For example, the shape of the receptacle **125** enables a user to fit the receptacle **125** over the hitch ball **220** using an easy place and drop motion, or to remove the receptacle **125** from the hitch ball **220** using an easy lift and remove motion. More specifically, the user can grasp the grab handle **110**, position the base section **105** over the trailer hitch ball **220**, and release grip of the grab handle **110** so that the single-part garbage can hitch **100** rests on the hitch ball **220** of a vehicle. A garbage can first be attached to the single-part garbage can hitch **100**, as further described below, and then the single-part garbage can hitch **100** can be placed onto to the hitch ball **220**. Alternatively, the single-part garbage can hitch **100** can be placed onto to the hitch ball **220** first, followed by the garbage can being attached to the single-part garbage can hitch **100**. The vehicle

can then tow the garbage can, as further described below. When finished towing the garbage can, the user can grasp the grab handle **110**, and lift the single-part garbage can hitch **100** off of the trailer hitch ball **220**, and then detach the single-part garbage can hitch **100** from the garbage can. Alternatively, the single-part garbage can hitch **100** can be detached from the garbage can first, followed by the lifting of the single-part garbage can hitch **100** off of the trailer hitch ball **220**.

(38) FIG. **3** illustrates a close-up side view of the single-part garbage can hitch **100** of FIGS. **1**A through **1**F with a handle bar **305** of a garbage can disposed in the semi-cylindrical well **115** in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **3** are described above, and therefore, a detailed description of these elements is not necessarily repeated. As can be seen in FIG. **3**, the handle bar **305** can fit within the semi-cylindrical well **115** sandwiched between the attachment arm **120** and an opposing wall **310** of the base section **105**. In other words, the attachment arm **120** can include a first sidewall **315** of the semi-cylindrical well **115**, and the base section **105** can include the opposing sidewall **310** opposite the first sidewall **315** of the semi-cylindrical well **115**. The handle bar **305** may freely twist within the semi-cylindrical well **115** as needed to secure the attachment arm **120** to the garbage can, as further explained below.

(39) FIG. **4** illustrates a perspective view of the single-part garbage can hitch **100** of FIGS. **1**A through **1**F being coupled with a garbage can **400** in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **4** are described above, and therefore, a detailed description of these elements is not necessarily repeated. The dashed lines shown in FIG. **4** are example boundaries of a garbage can **400**, but it will be understood that the relative dimensions of the garbage can **400** can be different without departing from the inventive concepts disclosed herein. The garbage can **400** can include a lid **405** and one or more sidewalls **410**.

(40) The attachment arm **120** of the single-part garbage can hitch **100** can slip over a center cross stiffener **415** of the garbage can **400**, thereby securing the single-part garbage can hitch **100** to the garbage can **400** in such a way that prevents significant lateral movement of the single-part garbage can hitch **100**. In other words, the single-part garbage can hitch **100** is prevented from sliding along the handle bar **305** of the garbage can **400**. Put differently, the center cross stiffener **415** can prevent the single-part garbage can hitch **100** from sliding sideways, such that the single-part garbage can hitch **100** can be kept substantially in a fixed position at or near a center region of the handle bar **305** of the garbage can **400**.

(41) More specifically, the recess **130** of the attachment arm **120** can be disposed in a horizontal section **420** of the attachment arm **120**, and can fit to the center cross stiffener **415**. A lip **430** of the horizontal section **420** of the attachment arm **120**, together with a vertical section **425** of the attachment arm **120**, can prevent the single-part garbage can hitch **100** from any significant lateral movement along the handle bar **305**. The dimensions of the recess **130** can be such that the attachment arm **120** fits snugly on the center cross stiffener **415**. The single-part garbage can hitch **100** can fit most standard sized garbage cans. Nevertheless, in some embodiments, the center cross stiffener **415** can be provided in different sizes depending on the garbage can manufacturer. Multiple types of single-part garbage can hitches **100** with varying dimensions of the recess **130** in order to fit and attach to any center cross stiffener **415** may be used. In some embodiments, the size of the recess **130** of the single-part garbage can hitch **100** is adjustable such that it can fit over or otherwise attach to any sized center cross stiffener **415**.

(42) In some embodiments, the grab handle **110** can include a first vertical sidewall **435**, a second vertical sidewall **440**, and a horizontal wall **445** coupled to the first vertical sidewall **435** and to the second vertical sidewall **440**. In some embodiments, the grab handle **110** is a material strap, a plastic strap, a rubber strap, or other suitable handle that is connected to the base section **105**.

(43) FIG. **5** illustrates a plan view of the single-part garbage can hitch **100** of FIGS. **1**A through **1**F being coupled with a trailer hitch ball **505** of a vehicle **510** in accordance with various

embodiments of the present inventive concept. Some of the reference numerals of FIG. **5** are described above, and therefore, a detailed description of these elements is not necessarily repeated. The dashed lines shown in FIG. **5** are example boundaries of a garbage can **400** and the vehicle **510**, but it will be understood that the relative dimensions of the garbage can **400** and the vehicle **510** can be different without departing from the inventive concepts disclosed herein. It will be understood that any type of vehicle **510** having a tow hitch ball can be used.

(44) Once the single-part garbage can hitch **100** is engaged with the garbage can **400**, it can be lowered onto the trailer hitch ball **505**, as shown at **515**. For example, the attachment arm **120** can be attached to the center cross stiffener **415** of the garbage can **400**, and the handle bar **305** of the garbage can **400** can rest in the semi-cylindrical well **115** of the single-part garbage can hitch **100**. Thereafter, the single-part garbage can hitch **100** can be grasped by the grab handle **110**, and moved together with garbage can **400** toward and then over the trailer hitch ball **505**. The weight of the garbage can **400** provides rigidity to the linkage, and once the single-part garbage can hitch **100** is attached to the hitch ball **500**, the weight of the garbage can **400** is transferred to the vehicle **510**. The load on the single-part garbage can hitch **100** itself is minimal. Because of the gravity-enhanced configuration providing downward levered pressure on the trailer hitch ball **505**, the single-part garbage can hitch **100** is advantageously kept securely linked to the trailer hitch ball **505**, even when the vehicle **510** drives over bumpy, gravelly, and/or dirt-based roads.

(45) FIG. **6** illustrates a side view of the single-part garbage can hitch **100** of FIGS. **1**A through **1**F coupling the garbage **400** can with the trailer hitch ball **505** of the vehicle **510** in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **6** are described above, and therefore, a detailed description of these elements is not necessarily repeated.

(46) The garbage can **400** can include one or more wheels **605**. After the garbage can **400** has been linked to the vehicle **510** using the single-part garbage can hitch **100**, the vehicle **510** can easily drag the garbage can **400** as the one or more wheels **605** of the garbage can **400** rotate on the ground. The weight of the garbage can **400** provides rigidity to the linkage, which keeps the single-part garbage can hitch **100** from coming off the hitch ball **505**. After the garbage can **400** has been dragged to the end of the drive way, for example, the single-part garbage can hitch **100** can be lifted from the hitch ball **505**. The single-part garbage can hitch **100** can then be easily detached from the garbage can **400** by disengaging the attachment arm **120**. Alternatively, the garbage can **400** can first be disengaged from the single-part garbage can hitch **100**, and then the single-part garbage can hitch **100** can be lifted from the hitch ball **505**. At no point does the garbage can **400** need to be lifted off the ground.

(47) FIG. **7** illustrates a perspective view of a garbage can **705** having a built-in hitch **700** for coupling the garbage can **705** with the trailer hitch ball (e.g., **505** of FIG. **5**) of the vehicle (e.g., **510** of FIG. **5**) in accordance with various embodiments of the present inventive concept. The built-in hitch **700** can be injection molded into the garbage can **705** itself. In an alternative embodiment, the built-in hitch **700** can be permanently attached to the garbage can **705**. In yet another embodiment, the built-in hitch **700** can be removably attached to the garbage can **705**. The built-in hitch **700** can simultaneously act as a stiffener for the handle bar **710** of the garbage can **705**. In some embodiments, the built-in hitch **700** is manufactured separately from the garbage can **705**, and then permanently or temporarily attached to the garbage can **705**. The attachment can be accomplished using screws, adhesives, welding, or the like.

(48) The built-in hitch **700** can include a base section **715**. In some embodiments, the base section **715** is substantially in the form of a block. However, it will be understood that the base section **715** can have other suitable shapes without departing from the inventive aspects described herein. The built-in hitch **700** can include an upper pedestal **720** having a semi-cylindrical well **730** disposed therein. The upper pedestal **720** and well **730** can receive, hold, and/or provide structural support to the handle bar **710**. In other words, the handle bar **710** can fit at least partially into, and be held at

least partially within, the semi-cylindrical well **730** of the upper pedestal **720**. In some embodiments, the handle bar **710** is free to rotate within the semi-cylindrical well **730** when a lid **735** of the garbage can **705** is opened or closed. In some embodiments, the handle bar **710** does not rotate, but is otherwise fixed or stationary within the well **730**. The garbage can **700** can include one or more sidewalls (e.g., **750**). The base section **715** can be attached to one of the sidewalls **750**.

(49) The built-in hitch **700** can include a receptacle **725**. The receptacle **725** can be located within the base section **715** of the built-in hitch **700**. The receptacle **725** can be shaped to receive a trailer hitch ball of a vehicle, as further described below. The garbage can **705** can include one or more wheels **740**. After the garbage can **705** has been linked to the vehicle (e.g., **510** of FIG. **5**) using the built-in hitch **700**, the vehicle **510** can easily drag the garbage can **705** as the one or more wheels **740** of the garbage can **705** rotate on the ground.

(50) FIG. **8** illustrates a plan view of the garbage can **705** having the built-in hitch **700** for coupling the garbage can **705** with the trailer hitch ball (e.g., **505** of FIG. **5**) of the vehicle (e.g., **510** of FIG. **5**) in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **8** are described above, and therefore, a detailed description of these elements is not necessarily repeated. The handle bar **710** can sit within the semi-cylindrical well **730** of the upper pedestal **720** of the built-in hitch **700** of the garbage can **705**. In some embodiments, the handle bar **710** is fixed or stationary within the well **730**. In some embodiments, the handle bar **710** can rotate within the well **730**. As can be seen in FIG. **8**, the built-in hitch **700** can take the place of a cross stiffener beam. In other words, the garbage can **705** need not have a cross stiffener beam because the built-in hitch can provide a dual function of structural support and the ability to attach to a ball hitch of a vehicle. In some embodiments, both a cross stiffener beam (e.g., **415** of FIG. **4**) and the built-in hitch **700** are part of the garbage can **705**.

(51) FIG. **9** illustrates a side view of the garbage can **705** having the built-in hitch **700** for coupling the garbage can **705** with the trailer hitch ball (e.g., **505** of FIG. **5**) of the vehicle (e.g., **510** of FIG. **5**) in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **9** are described above, and therefore, a detailed description of these elements is not necessarily repeated. A side view of the base section **715** and the pedestal **720** of the built-in hitch **700** is shown in FIG. **9**.

(52) FIG. **10** illustrates another side view of the garbage can **705** having the built-in hitch **700** for coupling the garbage can **705** with the trailer hitch ball (e.g., **505** of FIG. **5**) of the vehicle (e.g., **510** of FIG. **5**) in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **10** are described above, and therefore, a detailed description of these elements is not necessarily repeated. The side view in FIG. **10** shows the receptacle **725**, which can be placed over the trailer hitch (e.g., **505** of FIG. **5**).

(53) The receptacle **725** can include a hollow cylindrical section (e.g., **205** of FIG. **2**) and a hollow conical frustum section (e.g., **215** of FIG. **2**) having a tapered wall (e.g., **210** of FIG. **2**). The receptacle **725** can fit over any standard sized trailer hitch ball (e.g., **220**, **505**), such as 1⅞ inch, 2 inch, and 2 5/16 inch hitch balls. It will be understood that in alternate embodiments, the receptacle **725** can take other shapes that are suitable for being fit over any standard sized trailer hitch ball, without departing from the inventive aspects disclosed herein. For example, the receptacle **725** can have a hemispherical shape to receive the trailer hitch ball (e.g., **220**, **505**). In some embodiments, the receptacle **725** has a box shape, a hexagonal shape, a triangular shape, or the like.

(54) FIG. **11** illustrates yet another side view of the garbage can **705** having the built-in hitch **700** for coupling the garbage can **705** with the trailer hitch ball **505** of the vehicle **510** in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **11** are described above, and therefore, a detailed description of these elements is not necessarily repeated.

(55) As shown in FIG. **11**, the garbage can **705** having the built-in hitch **700** can be linked to the vehicle **510** for easy dragging. The garbage can **705** can include one or more wheels **740**. After the

garbage can **705** has been linked to the vehicle **510** using the built-in hitch **700**, the vehicle **510** can easily drag the garbage can **705** as the one or more wheels **740** of the garbage can **705** rotate on the ground. The weight of the garbage can **705** provides rigidity to the linkage, which keeps the built-in hitch **700** from coming off the hitch ball **505**. After the garbage can **705** has been dragged to the end of the drive way, for example, the garbage can **705** with the built-in hitch **700** can be lifted from the hitch ball **505**. At no point does the garbage can **705** need to be lifted off the ground. Because of the gravity-enhanced configuration providing downward levered pressure on the trailer hitch ball **505**, the built-in garbage can hitch **700** is advantageously kept securely linked to the trailer hitch ball **505**, even when the vehicle **510** drives over bumpy, gravelly, and/or dirt-based roads. The built-in hitch **700** provides the added advantage of not being a separate part, i.e., of not having the step of attaching the hitch **700** to the garbage can **705**, or removing the hitch **700** from the garbage can **705**, since the hitch **700** is built-in to the garbage can **705**.

(56) The built-in hitch **700** and the garbage can **705** can together be made of a single contiguous molded part. In an alternate embodiment, the built-in hitch **700** can be made of multiple separate parts that are combined (e.g., attached to each other), and attached to the garbage can **705**, to form a single functional part with the garbage can **705**. The built-in hitch **700** is preferably made of plastic because it is lightweight and strong, but it will be understood that the built-in hitch **700** can be made of any suitable hard material such as metal, fiber glass, wood, composite materials, or the like. In some embodiments, the built-in hitch **700** and/or the garbage can **705** are manufactured using a plastic injection mold. In some embodiments, the built-in hitch **700** and/or the garbage can **705** are manufactured using a 3D printer.

(57) FIG. **12** is a flow diagram **1200** illustrating a technique for using a single-part garbage can hitch to tow a garbage can using a vehicle. At **1205**, a single-part garbage can hitch can be attached to a garbage can. At **1210**, a receptacle of the single-part garbage can hitch can be placed over a hitch ball of a vehicle, thereby securing the garbage can to the vehicle. At **1215**, the garbage can may be towed using the single-part garbage can hitch and the vehicle.

(58) FIG. **13** is a flow diagram **1300** illustrating another technique for using the single-part garbage can hitch to tow the garbage can using the vehicle. At **1305**, a receptacle of the single-part garbage can hitch can be placed over the hitch ball of the vehicle. At **1310**, the single-part garbage can hitch can be attached to the garbage can, thereby securing the garbage can to the vehicle. At **1315**, the garbage can may be towed using the single-part garbage can hitch and the vehicle.

(59) FIG. **14** is a flow diagram **1400** illustrating a technique for manufacturing a single-part garbage can hitch to tow a garbage can using a vehicle. At **1405**, a mold can be created for a single-part garbage can hitch as described herein, e.g., having a base section, a grab handle, an attachment arm that is configured to attach to a garbage can, a receptacle disposed within the base section, and a semi-cylindrical well disposed within the base section. At **1410**, plastic or another suitable hardening substance can be injected into the mold. At **1415**, the single-part garbage can hitch can be removed from the mold.

(60) Alternatively, at **1420**, a design file of a single-part garbage can hitch as described herein can be loaded into a 3-D printer. At **1425**, the 3-D printer can print the single-part garbage can hitch having, e.g., a base section, a grab handle, an attachment arm that is configured to attach to a garbage can, a receptacle disposed within the base section, and a semi-cylindrical well disposed within the base section.

(61) FIG. **15** is a flow diagram **1500** illustrating a technique for manufacturing a garbage can having a built-in hitch to tow the garbage can using a vehicle. At **1505**, a mold can be created for a garbage can including a built-in hitch as described herein, e.g., having a base section, a receptacle disposed within the base section, a pedestal, and a semi-cylindrical well disposed within the pedestal. At **1515**, plastic or another suitable hardening substance can be injected into the mold. At **1520**, the garbage can include the built-in hitch can be removed from the mold.

(62) Alternatively, at **1520**, a design file of a garbage can including the built-in hitch can be loaded

into a 3-D printer. At **1525**, the 3-D printer can print the garbage can including, e.g., a built-in hitch as described herein, e.g., having a base section, a receptacle disposed within the base section, a pedestal, and a semi-cylindrical well disposed within the pedestal.

(63) FIG. **16** is a perspective view of a garbage can hitch **1600** removably attached to a garbage can **400** in accordance with various embodiments of the present inventive concept. FIG. **17** is a different perspective view of the garbage can hitch **1600** removably attached to the garbage can **400** in accordance with various embodiments of the present inventive concept. Reference is now made to FIGS. **16** and **17**.

(64) The garbage can **400** can include a lid **405** and one or more sidewalls **410**. A garbage can hitch **1600** comprises a base section **1610**, and a receptacle **1625** disposed within the base section **1610**. The garbage can hitch **1600** can receive a hitch ball (e.g., **505** of FIG. **5**) of a vehicle (e.g., **510** of FIG. **5**). In some embodiments, the garbage can **400** is attachable to the vehicle via the garbage can hitch **1600**, and the garbage can **400** can be dragged by the vehicle using the garbage can hitch **1600**.

(65) The garbage can hitch **1600** can include a semi-cylindrical well **1615** disposed within the base section **1610**. The semi-cylindrical well **1615** can receive a handle bar **305** of the garbage can **400**. The garbage can hitch **1600** can include a first arm **1630** that extends from the base section **1610**. The first arm **1630** can be positioned adjacent a first side of center cross stiffener **415** of the garbage can **400**. The garbage can hitch **1600** can include a second arm **1635** that extends from the base section **1610**. The second arm **1635** can be positioned adjacent a second side of the center cross stiffener **415** of the garbage can **400**.

(66) In some embodiments, the garbage can hitch **1600** includes a first width-adjusting clamp pad **1640**, which can be removably attached to the first arm **1630**. The garbage can hitch **1600** can include a second width-adjusting clamp pad **1645**, which can be removably attached to the second arm **1635**. The first width-adjusting clamp pad **1640** and the second width-adjusting clamp pad **1645** can be clamped against the center cross stiffener **415** of the garbage can **400** to hold the garbage can hitch **1600** in place. The width-adjusting clamp pads (e.g., **1640**, **1645**) can be made of plastic, rubber, foam, or other suitable material that can be clamped against the center cross stiffener **415** of the garbage can **400**, and hold the garbage can hitch **1600** to the garbage can **400**. The first width-adjusting clamp pad **1640** can be press-fit to the first arm **1630**, as further described below. The second width-adjusting clamp pad **1645** can be press-fit to the second arm **1635**, as also further described below.

(67) It will be understood that the width-adjusting clamp pads can be removably attached to the first and second arm in a manner that is not necessarily press-fit. For example, a width-adjusting clamp pad can be slid into an arm of the garbage can hitch. By way of another example, a width-adjusting clamp pad can be locked into an arm of the garbage can hitch. By way of yet another example, a width-adjusting clamp pad can be slid and locked into an arm of the garbage can hitch. By way of still another example, a width-adjusting clamp pad can be snapped into an arm of the garbage can hitch. By way of yet another example, a width-adjusting clamp pad can be latched to an arm of the garbage can hitch.

(68) FIG. **18** is another perspective view of the garbage can hitch **1600** removably attached to the garbage can **400** in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **18** are described above, and a detailed description of these is not necessarily repeated. Reference is now made to FIGS. **16-18**.

(69) In some embodiments, the garbage can hitch **1600** can include a first opening **1648** through the first arm **1630**, as shown in FIGS. **16** and **17**. The garbage can hitch **1600** can include a second opening **1805** through the second arm **1635**, as shown in FIG. **18**. The garbage can hitch **1600** can include a fastener **1650**, as shown in FIGS. **16** and **17**, that can be disposed through the first opening **1648** and the second opening **1805** to keep the handle bar **305** of the garbage can **400** seated within the semi-cylindrical well **1615** of the base section **1610**. In other words, the fastener

1650 can prevent the garbage can hitch **1600** from falling off of the garbage can **400** due to the forces of gravity. The fastener **1650** need not be permanently attached to the garbage can **400**. The fastener **1650** can be comprised of a metal rod, a wooden rod, a plastic rod, or the like. In some embodiments, the fastener **1650** is in a different shape such as a rectangle. In some embodiments, the fastener **1650** is a clip, a wire, or the like.

(70) In some embodiments, the fastener **1650** can be disposed through the first width-adjusting clamp pad **1640** and/or the second width-adjusting clamp pad **1645**. The garbage can hitch **1600** can include a third opening **1810** through the second arm **1635**. The garbage can hitch **1600** can include a second fastener **1815** that can be disposed through the third opening **1810**. The second fastener **1815** can be the same type or similar type as the first fastener **1650**. In some embodiments, the second fastener **1815** can be in contact with the center cross stiffener **415** of the garbage can **400**. In some embodiments, the second fastener **1815** can include drill threads, and can be drilled into the center cross stiffener **415** of the garbage can **400**. The second fastener **1815** can provide additional stability and stiffness to the attachment of the garbage can hitch **1600** to the garbage can **400**.

(71) In some embodiments, the receptacle **1625** includes a hollow cylindrical section. In some embodiments, the receptacle **1625** includes a hollow conical frustum section having a tapered wall. In some embodiments, the base section **1610**, the first arm **1630**, the second arm **1635**, and the receptacle **1625** form a single contiguous part. In some embodiments, the single contiguous part is an injection molded single part made of plastic.

(72) FIG. **19** is a close-up view of the garbage can hitch **1600** removably attached to the garbage can **400** in accordance with various embodiments of the present inventive concept. FIG. **20** is another close-up view of the garbage can hitch **1600** removably attached to the garbage can **400** in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIGS. **19** and **20** are described above, and a detailed description of these is not necessarily repeated. Reference is now made to FIGS. **16-20**.

(73) The first arm **1630** can include a first slot **1905**. The second arm **1635** can include a second slot **1910**. The first width-adjusting clamp pad **1640** can include a first ridge **1915**. The second width-adjusting clamp pad **1645** can include a second ridge **1920**. The first ridge **1915** of the first width-adjusting clamp pad **1640** can be press-fit into the first slot **1905** of the first arm **1630**. The second ridge **1920** of the second width-adjusting clamp pad **1645** can be press-fit into the second slot **1920** of the second arm **1635**. The first ridge **1915** of the first width-adjusting clamp pad **1640** can be removed from the first slot **1905** of the first arm **1630**. The second ridge **1920** of the second width-adjusting clamp pad **1645** can be removed from the second slot **1920** of the second arm **1635**. It will be understood that the ridges (e.g., **1915**, **1920**) can be of a different shape, such as a wedge, a semi-cylinder, a hook, or the like.

(74) FIG. **21** is a perspective view of the garbage can hitch **1600** including various different width-adjusting clamp pads (e.g., **1640**, **1645**, **2140***a*, **2145***a*, **2140***b*, **2145***b*, **2140***c*, and **2145***c*) in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **21** are described above, and a detailed description of these is not necessarily repeated. Reference is now made to FIGS. **16-21**.

(75) The garbage can hitch **1600** can include an opening **2105** between the first arm **1630** and the second arm **1635**, which can accommodate the center cross stiffener **415** of the garbage can **400**. In some embodiments, the first width-adjusting clamp pad **1640** can be swapped for a third width-adjusting clamp pad (e.g., **2140***a*) having a different width than the first width-adjusting clamp pad **1640**. The second width-adjusting clamp pad **1645** can be swapped for a fourth width-adjusting clamp pad (e.g., **2145***a*) having a different width than the second width-adjusting clamp pad **1645**. In other words, any of the width-adjusting clamp pads (e.g., **1640**, **1645**, **2140***a*, **2145***a*, **2140***b*, **2145***b*, **2140***c*, and **2145***c*) can be swapped for any of the other width-adjusting clamp pads until a suitable width of a gap between the two installed width-adjusting clamp pads is achieved. The

suitable width of the gap between the two installed width-adjusting clamp pads is one that accommodates and/or presses against the center cross stiffener **415** of the garbage can **400**. In other words, the suitable width of the gap between the two installed width-adjusting clamp pads can be one that causes the width-adjusting clamp pads to clamp against the center cross stiffener **415** of the garbage can **400** to hold the garbage can hitch **1600** to the garbage can **400**.

(76) Even though only one of the width-adjusting clamp pads **2140***b*, **2145***b*, is shown in FIG. **21**, it will be understood that there may be two width-adjusting clamp pads **2140***b*, **2145***b* of the same dimension available for insertion into either arm and/or both arms (e.g., **1630**, **1635**) of the garbage can hitch **1600**. Similarly, even though only one of the width-adjusting clamp pads **2140***c*, **2145***c*, is shown in FIG. **21**, it will be understood that there may be two width-adjusting clamp pads **2140***c*, **2145***c* of the same dimension available for insertion into either arm and/or both arms (e.g., **1630**, **1635**) of the garbage can hitch **1600**.

(77) FIG. **22** is a schematic diagram of side views of the various different width-adjusting clamp pads in accordance with various embodiments of the present inventive concept. As can be seen in FIG. **22**, a suitable width (e.g., ¼ inch, 1⅛ inch, 2⅛ inch, 3⅛ inch) of a gap **2205** between the width-adjusting clamp pads (e.g., **1640**, **1645**, **2140***a*, **2145***a*, **2140***b*, **2145***b*, **2140***c*, and **2145***c*) can be created by insertion of the width-adjusting clamp pads (e.g., **1640**, **1645**, **2140***a*, **2145***a*, **2140***b*, **2145***b*, **2140***c*, and **2145***c*) into the arms (e.g., **1630**, **1635**) of the garbage can hitch **1600**. The suitable widths shown in FIG. **22** are designed to fit with most center cross stiffeners (e.g., **415**) of most garbage cans (e.g., **400**); however, it will be understood that other sizes of width-adjusting clamp pads can be used, and other suitable widths provided.

(78) FIG. **23** is a bottom view of the garbage can hitch **1600** including the receptacle **1625** in accordance with various embodiments of the present inventive concept. FIG. **24** is a top view of the garbage can hitch **1600** in accordance with various embodiments of the present inventive concept. FIG. **25** is a first side view of the garbage can hitch **1600** in accordance with various embodiments of the present inventive concept. FIG. **26** is a second side view of the garbage can hitch **1600** in accordance with various embodiments of the present inventive concept. Some of the reference numerals of FIG. **23** are described above, and a detailed description of these is not necessarily repeated. The dimensions shown are example dimensions, are such that they are optimal for fitting most types of garbage cans. However, it will be understood that other dimensions can be used without departing from the inventive concepts disclosed herein.

(79) Some embodiments include a garbage can hitch, comprising a base section, a grab handle coupled to the base section, and an attachment arm coupled to the base section, and configured to attach to a garbage can. The garbage can hitch can include a receptacle disposed within the base section, and configured to receive a hitch ball of a vehicle. The garbage can hitch can include a semi-cylindrical well disposed within the base section, and configured to receive a handle bar of the garbage can.

(80) In some embodiments, the attachment arm includes a first sidewall of the semi-cylindrical well, and the base section includes a second sidewall opposite the first sidewall of the semi-cylindrical well. In some embodiments, the receptacle includes a hollow cylindrical section. In some embodiments, the receptacle includes a hollow conical frustum section having a tapered wall. In some embodiments, the attachment arm includes a recess located at an underside thereof. In some embodiments, the attachment arm includes a vertical section and a horizontal section. In some embodiments, the horizontal section includes a lip. In some embodiments, the recess of the attachment arm and the lip are configured to be attached to a cross stiffener of the garbage can to prevent lateral movement of the garbage can hitch along a handle bar of the garbage can.

(81) In some embodiments, the grab handle includes a first vertical sidewall, a second vertical sidewall, and a horizontal wall coupled to the first vertical sidewall and to the second vertical sidewall. In some embodiments, the base section, the attachment arm, and the receptacle are arranged in a gravity-enhanced configuration to provide downward levered pressure on the hitch

ball responsive to the garbage can being removably attached to the vehicle. In some embodiments, the garbage can is attachable to the vehicle via the garbage can hitch, and the garbage can is configured to be dragged by the vehicle using the garbage can hitch. In some embodiments, the base section, the grab handle, the attachment arm, and the receptacle form a single contiguous part. In some embodiments, the single contiguous part is an injection molded single part made of plastic.

(82) Some embodiments include a garbage can, comprising one or more sidewalls, and a built-in hitch attached to the one or more sidewalls. In some embodiments, the built-in hitch includes a base section, and a receptacle disposed within the base section, and configured to receive a hitch ball of a vehicle. The garbage can may further include a handle bar. The built-in hitch can include a pedestal coupled to the base section of the built-in hitch. The built-in hitch can include a semi-cylindrical well disposed within the pedestal of the built-in hitch, wherein the handle bar is held at least partially within the semi-cylindrical well.

(83) In some embodiments, the receptacle includes a hollow cylindrical section. In some embodiments, the built-in hitch is attachable to the vehicle, and the garbage can is configured to be dragged by the vehicle using the built-in hitch. In some embodiments, the one or more sidewalls of the garbage can and the built-in hitch form a single contiguous part. In some embodiments, the single contiguous part is an injection molded single part made of plastic.

(84) In some embodiments, a garbage can hitch comprises a base section, and a receptacle disposed within the base section, and configured to receive a hitch ball of a vehicle. The garbage can hitch can include a semi-cylindrical well disposed within the base section, and configured to receive a handle bar of a garbage can. The garbage can hitch can include a first arm that extends from the base section. The first arm can be positioned adjacent a first side of center cross stiffener of the garbage can. The garbage can hitch can include a second arm that extends from the base section. The second arm is configured to be positioned adjacent a second side of the center cross stiffener of the garbage can.

(85) In some embodiments, the garbage can hitch includes a first width-adjusting clamp pad that is configured to be removably attached to the first arm. The garbage can hitch can include a second width-adjusting clamp pad that is configured to be removably attached to the second arm. The first width-adjusting clamp pad and the second width-adjusting clamp pad can be configured to be clamped against the center cross stiffener of the garbage can. The first width-adjusting clamp pad can be configured to be press-fit to the first arm. The second width-adjusting clamp pad can be configured to be press-fit to the second arm.

(86) In some embodiments, the first width-adjusting clamp pad is configured to be swapped for a third width-adjusting clamp pad having a different width than the first width-adjusting clamp pad. The second width-adjusting clamp pad can be configured to be swapped for a fourth width-adjusting clamp pad having a different width than the second width-adjusting clamp pad. The first arm can include a first slot. The second arm can include a second slot. The first width-adjusting clamp pad can include a first ridge. The second width-adjusting clamp pad can include a second ridge. The first ridge of the first width-adjusting clamp pad can be configured to be press-fit into the first slot of the first arm. The second ridge of the second width-adjusting clamp pad can be configured to be press-fit into the second slot of the second arm.

(87) In some embodiments, the garbage can hitch can include a first opening through the first arm. The garbage can hitch can include a second opening through the second arm. The garbage can hitch can include a fastener that is configured to be disposed through the first opening and the second opening to keep the handle bar of the garbage can seated within the semi-cylindrical well of the base section.

(88) In some embodiments, the garbage can hitch can include a first width-adjusting clamp pad that is configured to be removably attached to the first arm. The garbage can hitch can include a second width-adjusting clamp pad that is configured to be removably attached to the second arm. The fastener can be configured to be disposed through the first width-adjusting clamp pad and the

second width-adjusting clamp pad. In some embodiments, the fastener is a first fastener. The garbage can hitch can include a third opening through the second arm. The garbage can hitch can include a second fastener that is configured to be disposed through the third opening.
(89) In some embodiments, the second fastener is configured to be contact with the center cross stiffener of the garbage can. In some embodiments, the second fastener is configured to be drilled into the center cross stiffener of the garbage can. In some embodiments, the garbage can is attachable to the vehicle via the garbage can hitch, and the garbage can is configured to be dragged by the vehicle using the garbage can hitch.
(90) In some embodiments, the receptacle includes a hollow cylindrical section. In some embodiments, the receptacle includes a hollow conical frustum section having a tapered wall. In some embodiments, the base section, the first arm, the second arm, and the receptacle form a single contiguous part. In some embodiments, the single contiguous part is an injection molded single part made of plastic.
(91) Having described and illustrated the principles of the inventive concept with reference to illustrated embodiments, it will be recognized that the illustrated embodiments can be modified in arrangement and detail without departing from such principles, and can be combined in any desired manner. And although the foregoing discussion has focused on particular embodiments, other configurations are contemplated. In particular, even though expressions such as “according to an embodiment of the invention” or the like are used herein, these phrases are meant to generally reference embodiment possibilities, and are not intended to limit the inventive concept to particular embodiment configurations. As used herein, these terms can reference the same or different embodiments that are combinable into other embodiments.
(92) Consequently, in view of the wide variety of permutations to the embodiments described herein, this detailed description and accompanying material is intended to be illustrative only, and should not be taken as limiting the scope of the inventive concept. What is claimed as the invention, therefore, is all such modifications as may come within the scope and spirit of the following claims and equivalents thereto.

Claims

1. A garbage can hitch, comprising: a base section; a receptacle disposed within the base section, and configured to receive a hitch ball of a vehicle; a semi-cylindrical well disposed within the base section, and configured to receive a handle bar of a garbage can; a first arm that extends from the base section, wherein the first arm is configured to be positioned adjacent a first side of a center cross stiffener of the garbage can; a second arm that extends from the base section, wherein the second arm is configured to be positioned adjacent a second side of the center cross stiffener of the garbage can; a first width-adjusting clamp pad that is configured to be removably attached to the first arm; and a second width-adjusting clamp pad that is configured to be removably attached to the second arm, wherein the first width-adjusting clamp pad and the second width-adjusting clamp pad are configured to be clamped against the center cross stiffener of the garbage can.
2. The garbage can hitch of claim 1, wherein: the first width-adjusting clamp pad is configured to be press-fit to the first arm; and the second width-adjusting clamp pad is configured to be press-fit to the second arm.
3. The garbage can hitch of claim 1, wherein: the first width-adjusting clamp pad is configured to be swapped for a third width-adjusting clamp pad having a different width than the first width-adjusting clamp pad; and the second width-adjusting clamp pad is configured to be swapped for a fourth width-adjusting clamp pad having a different width than the second width-adjusting clamp pad.
4. The garbage can hitch of claim 1, wherein: the first arm includes a first slot; and the second arm includes a second slot.

5. The garbage can hitch of claim 4, wherein: the first width-adjusting clamp pad includes a first ridge; and the second width-adjusting clamp pad includes a second ridge.
6. The garbage can hitch of claim 5, wherein: the first ridge of the first width-adjusting clamp pad is configured to be press-fit into the first slot of the first arm; and the second ridge of the second width-adjusting clamp pad is configured to be press-fit into the second slot of the second arm.
7. The garbage can hitch of claim 1, further comprising: a first opening through the first arm; a second opening through the second arm; and a fastener that is configured to be disposed through the first opening and the second opening to keep the handle bar of the garbage can seated within the semi-cylindrical well of the base section.
8. The garbage can hitch of claim 7, further comprising: a first width-adjusting clamp pad that is configured to be removably attached to the first arm; and a second width-adjusting clamp pad that is configured to be removably attached to the second arm, wherein the fastener is configured to be disposed through the first width-adjusting clamp pad and the second width-adjusting clamp pad.
9. The garbage can hitch of claim 7, wherein the fastener is a first fastener, the garbage can hitch further comprising: a third opening through the second arm; and a second fastener that is configured to be disposed through the third opening.
10. The garbage can hitch of claim 9, wherein the second fastener is configured to be contact with the center cross stiffener of the garbage can.
11. The garbage can hitch of claim 10, wherein the second fastener is configured to be drilled into the center cross stiffener of the garbage can.
12. The garbage can hitch of claim 1, wherein the garbage can is attachable to the vehicle via the garbage can hitch, and the garbage can is configured to be dragged by the vehicle using the garbage can hitch.
13. The garbage can hitch of claim 1, wherein: the base section, the first arm, the second arm, and the receptacle form a single contiguous part; and the single contiguous part is an injection molded single part made of plastic.
14. A garbage can hitch, comprising: a base section; a receptacle disposed within the base section, and configured to receive a hitch ball of a vehicle; and a semi-cylindrical well disposed within the base section, and configured to receive a handle bar of a garbage can, wherein: the receptacle includes a hollow cylindrical section; and the receptacle includes a hollow conical frustum section having a tapered wall.
15. A garbage can hitch, comprising: a base section; a receptacle disposed within the base section, and configured to receive a hitch ball of a vehicle; a semi-cylindrical well disposed within the base section, and configured to receive a handle bar of a garbage can; a first arm that extends from the base section, wherein the first arm is configured to be positioned adjacent a first side of center cross stiffener of the garbage can; a second arm that extends from the base section, wherein the second arm is configured to be positioned adjacent a second side of the center cross stiffener of the garbage can; a first opening through the first arm; a second opening through the second arm; and a fastener that is configured to be disposed through the first opening and the second opening to keep the handle bar of the garbage can seated within the semi-cylindrical well of the base section.